Ohm’s law

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect voltage, current and resistance for an ohmic component.

SUCCESS CRITERIA - I can...

1. Identify the voltage across a resistor.
2. Use V=IR with appropriate units.
3. Predict changes while stating what stays fixed.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Move from r to 3 r away from a point charge. Compare field magnitude and state the source model assumption.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: V = IR; I = V/R.
Ohmic resistor under fixed physical conditions. Resistance can change with temperature.

Worked reasoning: Across a 4 ohm resistor, a 12 V supply gives I=V/R=3 A. Doubling resistance at the same voltage halves current.

Your turn: Draw a one-resistor circuit, label supply voltage and current, and mark both resistor terminals.

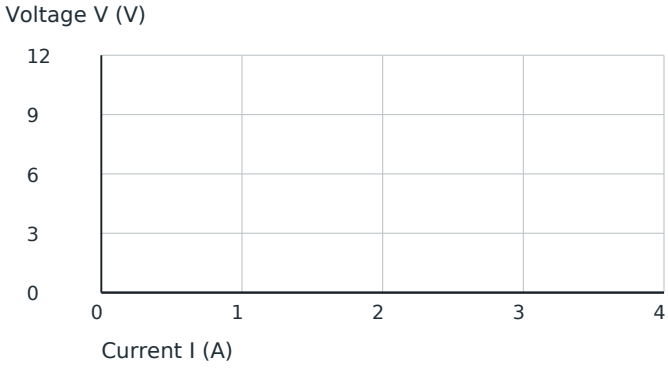


Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

03 Read and interpret the model

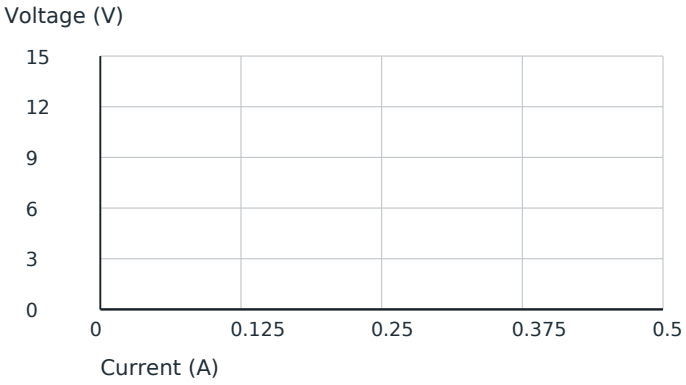
Sketch a graph of V against I for a fixed ohmic resistor. What does its slope represent?



Marzano check - circle: ① ② ③

04 Apply the physics

A resistor has 12 V across its terminals and carries 0.40 A. Find resistance. If the voltage becomes 9 V and resistance stays constant, find the new current. Label both operating points on the graph.



Marzano check - circle: ① ② ③

05 Explain your reasoning

A student doubles resistance while keeping voltage fixed and predicts twice the current. Correct the prediction.

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

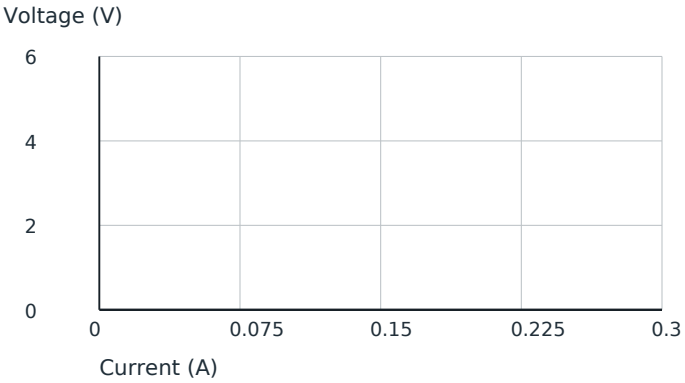
An EM wave in vacuum has frequency 6×10^{9} Hz. Use $c=3.00\times10^{8}$ m/s. Find wavelength and name the spectral region.
Label a wave sketch to support the calculation or comparison.



Marzano check - circle: ① ② ③

07 Exit check

Measured values are (I, V)=(0.10, 2.0),(0.20, 4.0),(0.30, 6.0). Find resistance and explain what supports an ohmic model.



Marzano check - circle: 1 2 3

08 STRETCH: test the model

An ohmic component has 4 V across it at 0.5 A. Find resistance and current at 10 V. Explain why this extrapolation requires the component to remain ohmic.



Label component values, current and voltage as needed.

Marzano check - circle: 1 2 3